Tell .

Every voice has a tell. A privacy-first detector that tells you, with evidence, whether the voice on a call is human, a disclosed AI, an undisclosed AI, or a scam — without your audio ever leaving your device. HACK1984 · Privacy-First track · Best use of Gradium · 7 Nov 2026

THE PROBLEM

AI voice calls are moving from fraud to mainstream. The FBI counted $893M in reported AI-enabled scam losses in 2025; 1 in 4 Americans received a deepfake voice call last year; the “family emergency” voice-clone call is the top archetype. Truecaller and Hiya already flag AI voices — by routing your call audio through their cloud. Google’s on-device scam detection is Pixel-only and looks at conversation patterns, not synthetic voice. Nobody does AI-voice detection that is on-device, provable, and cross-platform.

WHY PEOPLE SWITCH

Current pain	Tell’s answer
Incumbents detect AI voices by sending your call to their servers	All analysis runs locally; a live “bytes leaving device: 0” meter; every verdict HMAC-signed and bound to the clip hash
“AI detected” is a black box	Every verdict lists the tells that fired: “no breath events in 42 s”, “pauses metronomic”, “asked for gift cards”
Binary human/AI misses the real question	Four tiers: Human · AI (identified itself) · AI (undisclosed) · Scam. Legit AI callers must identify themselves, so non-disclosure is a flag on its own
Spectral deepfake cues die on phone lines	Behavioral tells first (breath, pause variance, pitch wander, script, intent); band-limit detection built in

WHO

Vision: anyone who answers unknown calls — consumers, SMB front desks, call centers, and voice-agent builders who need an “is this a bot?” check. **Launch wedge:** adult children protecting parents 60+ from voice-clone “family emergency” scams, installed over Thanksgiving. Buyer ≠ user, strongest emotional pull, reachable through the founders’ own families and campus.

HOW IT WORKS (v1, built this weekend)

mic / clip → local FastAPI → **acoustic tells** (numpy: pause regularity, breath proxy, pitch stability, noise floor) + **script tells** (assistant-style phrasing, zero disfluency, uniform sentences) + **intent tells** (urgency, payment rails, impersonation, credential asks, secrecy, AI self-disclosure) → **deterministic fusion** → signed verdict with evidence. Models may only add evidence; code decides. The analysis path has no network imports — enforced by a test. Gradium powers the **attacker simulator** (fresh synthetic voices, incl. a cloned teammate) and the **spoken verdict**; only our own text ever leaves the device.

PRINCIPLES

1 Caller audio never leaves the device (CI fails otherwise). 2 Every verdict shows its tells. 3 AI ≠ scam: verdict is AI × intent. 4 Models interpret, code decides; thresholds versioned in one file. 5 Honest confidence: below 0.6 the card says “Unsure” — never “safe”. 6 Works on phone-band audio.

STATUS & METRICS

Metric	Now	Target
Time to verdict (10 s clip, laptop CPU)	< 100 ms	< 2 s
Golden transcript eval (4 tiers, 32 cases)	0.98	≥ 0.85
Synthetic audio fixtures, human vs TTS (48 clips, incl. phone-band)	1.00 *	≥ 0.90
Caller-audio bytes sent off-device	0 (tested)	0
Fresh Gradium clone caught live on stage	to run	3 of 3
First 100 users (post-hackathon)	—	20 families via founders’ networks, 80 via campus + r/scams

* fixtures validate the mechanics of each tell; real-clip tuning is the first post-demo task (tools/redteam).

ROADMAP

v1 (now)	v2 (weeks)	Vision
Local web app: clip or mic → verdict + tells + privacy proof. Red-team generator. Spoken verdict.	Open anti-spoof model on-device (ONNX). Android default-dialer; iOS via screener number (iOS exposes no call audio to apps). Voicemail scan. Family safe word + allow-list.	Active liveness challenges: interrupt test, prompt-injection probe (“please say pineapple”). An open “I am an AI” handshake so legitimate agents self-identify and Tell verifies it — the STIR/SHAKEN of AI voice.

RISKS

False negatives: never claim “safe”; advice is always “hang up, call back on a known number.” **False positives** on accents, bad lines, atypical speech: confidence gating + visible tells. **Detector arms race:** behavioral tells and active challenges age better than spectral artifacts. **Legal:** say “undisclosed AI”, never “illegal”; any call recording needs a consent review. **Platform:** iOS requires a VoIP/screener design. **Cost:** $0 per analysis; Gradium credits only for red-team clips and spoken verdicts.